

# On the Cryptographic Resilience of MDS Matrices

Kamil Otal<sup>1</sup>, Ali Mert Sülçe<sup>2</sup>, and Oğuz Yayla<sup>2</sup>

<sup>1</sup> TÜBİTAK BİLGEM National Research Institute of Electronics and Cryptology  
Gebze, Kocaeli, Türkiye  
kamil.otal@gmail.com

<sup>2</sup> Institute of Applied Mathematics, Middle East Technical University  
Çankaya, Ankara, Türkiye  
{mert.sulce, oguz}@metu.edu.tr

**Abstract.** The zero-difference attack on AES, introduced by Barddeh and Rijmen in [ToSC 2022(2):43–62], exploits some structural properties -referred to as related differentials- in the AES MDS matrix. Daemen and Rijmen earlier demonstrated that these related differentials appear not only in the AES MixColumns matrix but in all  $4 \times 4$  circulant MDS matrices [CCDS 2009(1):47–69]. In the same paper, they also showed an example of  $4 \times 4$  Hadamard MDS matrices for which there exists no related differentials. Combining both results, we can say for example that some  $4 \times 4$  Hadamard MDS matrices are more “secure” than any  $4 \times 4$  circulant MDS matrices.

Recently, Jha et al. investigated whether it is possible to characterize  $4 \times 4$  Hadamard MDS matrices for which we can find no related differentials in [IACR Commun. Cryptol. 2(1): 37 (2025)]. As a result, they gave a systematic method to construct such “secure” matrices with respect to their parameters.

In this paper, we investigate the same problem for all  $2 \times 2$  and  $3 \times 3$  matrices to understand the cryptographic resilience of MDS matrices both theoretically and practically. As a result, we obtain the following results:

- There are no related differentials for any  $2 \times 2$  MDS matrices.
- There exist related differentials for all  $3 \times 3$  circulant MDS matrices.
- There exist related differentials for all  $3 \times 3$  involutory MDS matrices when the finite field has even size.
- We characterize all  $3 \times 3$  MDS matrices for which there exist no related differentials when the size of the finite field is even.

In this way, we fill a gap for the cryptographic resilience problem of MDS matrices over finite fields.

**MSC 2020:** 94A60, 11T71.

**Keywords:** Maximum distance separable (MDS) matrices · Related differentials · Zero-difference attack

## 1 Introduction

The Advanced Encryption Standard (AES) [AES01] has been intensely utilized and analyzed all over the world since the day it was standardized. In the construction of a modern block cipher like AES, there are two main principles taken into account: confusion and diffusion. The confusion property is generally provided by substitution-boxes (S-boxes) which are special nonlinear mappings on a finite dimensional vector space over a finite field  $\mathbb{F}_q$  of  $q$  elements, whereas the diffusion is often provided by a maximum distance separable (MDS) matrix which is the redundancy (check) part of a generator matrix of an MDS code.

Specifically, if  $r$ -bit to  $r$ -bit S-boxes are chosen during the construction of a block cipher, then an MDS matrix over a finite field  $\mathbb{F}_{2^r}$  (or  $\text{GL}(r, \mathbb{F}_2)$  in general) is preferred for the maximum possible diffusion, see the wide trail strategy approach in [AES01] for further detail. Such a preference lets us easily count active S-boxes through rounds. Note that counting active S-boxes is an important process to show the resistance of the cipher against linear [Mat94] and differential cryptanalysis [BS91].

### 1.1 Related Work

MDS matrices have a special place in cryptography by considering this maximum possible branching feature. In recent decades, plenty of studies examined the construction of efficient MDS matrices and their implementations.

In addition to efficiency concerns, there are security analyzes of MDS matrices. Daemen and Rijmen [DR09] introduced the notion of related differentials for MDS matrices and showed examples of MDS matrices with and without related differentials. Later, Ronjom et al. [RBH17] presented the notion of zero-difference patterns which are deterministic 4-round characteristics for AES and a new key recovery attack on 5-round AES. Lastly, Bardeh and Rijmen [BR22] provided a general formulation of the zero-difference property which allows to combine the 4-round zero-difference property with related differentials and hence obtained a new 7-round related differentials characteristic, as a result they presented the first key-recovery attack on 7-round AES-128 based on the zero-difference property.

Lastly, Jha et al. investigated whether it is possible to characterize  $4 \times 4$  Hadamard MDS matrices for which we can find no related differentials in [JLG25]. As a result, they gave a systematic method to construct such “secure” matrices with respect to their parameters.

We would like to emphasize that, among several attack types, there were only two attacks, namely meet-in-the-middle (see for example [DS08, DKS10, DFJ13]) and impossible differential (see for example [LDKK08, BLNS18, LP21]) attacks, which have achieved to break the longest round-reduced versions of AES (e.g. up to 7 rounds of AES-128) except the 7-round attack on AES-128 of Bardeh and Rijmen in [BR22].

## 1.2 Motivation

We remark that zero-difference attacks also seem applicable for most of the SPN ciphers in the literature. Note also that, in general, MDS matrices have been assumed to have equivalent cryptographic resilience and hence cryptographers have considered solely their efficiency while examining such matrices. It is a very new and rare effort to examine MDS matrices from the security perspective but it seems that such efforts will intensify.

It is meaningful to investigate such “secure” MDS matrices for  $4 \times 4$  matrices as a first step since such dimensions are more widely used. However, there are  $3 \times 3$ ,  $8 \times 8$ ,  $16 \times 16$ ,  $24 \times 24$ , and  $32 \times 32$  MDS matrices in use in cryptography (see, for example, [G<sup>+</sup>21,BNN<sup>+</sup>10,SLS<sup>+</sup>23,AMPŠ24]). Therefore, investigation of such dimensions is interesting because of their not only theoretical importance but also practical roles.

## 1.3 Our Contribution

In this paper, we provide an investigation of existence of related differentials for  $2 \times 2$  and  $3 \times 3$  MDS matrices. We provide the following results.

- There are no related differentials for any  $2 \times 2$  MDS matrices.
- There exist related differentials for all  $3 \times 3$  circulant MDS matrices.
- There exist related differentials for all  $3 \times 3$  involutory MDS matrices when the finite field has even size.
- We characterize all  $3 \times 3$  MDS matrices for which there exist no related differentials when the size of finite field is even.

We emphasize that an arbitrary  $3 \times 3$  matrix has 9 independent entries as well as  $4 \times 4$  Hadamard (or circulant) matrices have 4 independent entries. That makes the investigation of all  $3 \times 3$  MDS matrices harder at first sight. We therefore apply a suitable equivalence idea and hence reduce the number of independent entries to 4 while we characterize all  $3 \times 3$  matrices.

## 1.4 Organization of the Paper

The following section presents some preliminary information to make the paper as self-contained as possible. In Section 3 we prove that there are no related differentials for any  $2 \times 2$  MDS matrices. Section 4 contains the proof of existence of related differentials for all  $3 \times 3$  circulant MDS matrices. A similar result for all  $3 \times 3$  involutory MDS matrices when the finite field has even characteristic is available in Section 5. Section 6 is devoted to the proof of the existence of related differentials for any  $3 \times 3$  MDS matrices when the finite field has even size. Lastly, Section 7 concludes the paper and presents some questions as possible future work.

## 2 Preliminaries

This section presents some preliminary information in two subsections: MDS matrices and related differentials.

### 2.1 MDS Matrices

In this section, we briefly recall the fundamental notions of MDS matrices, mostly coming from coding theory.

Let  $\mathbb{F}_q^n$  denote the set of vectors of length  $n$  and with entries from the finite field  $\mathbb{F}_q$  of size  $q$ . The function  $d : \mathbb{F}_q^n \times \mathbb{F}_q^n \rightarrow \mathbb{R}$  given by  $d(u, v) := |\{i : i \in \{1, 2, \dots, n\}, u_i \neq v_i\}|$  satisfies the metric properties and is called the *Hamming distance*. A subset  $C$  of  $\mathbb{F}_q^n$  endowed with the Hamming metric is called a *code*. We define the *minimum (Hamming) distance* of a code  $C$  by  $d(C) := \min\{d(u, v) : u, v \in C, u \neq v\}$ . There is an upper bound on  $d(C)$  given by  $d(C) \leq n + 1 - \log_q(|C|)$  and called the *Singleton bound*. A subset  $C$  of  $\mathbb{F}_q^n$  satisfying the Singleton bound is called an *MDS code*.

The main parameters of a code  $C$  in  $\mathbb{F}_q^n$  is denoted by  $(n, |C|, d(C))_q$ . In particular, if  $C$  is a  $k$ -dimensional subspace of  $\mathbb{F}_q^n$  over  $\mathbb{F}_q$ , then we denote the main parameters of  $C$  by  $[n, k, d(C)]_q$ . MDS codes exist for many but not all parameters. In particular, the *MDS Conjecture*, which is stated below, formulates the existence of linear MDS codes with respect to their parameters (see also [Rot06, Conjecture 11.16] for example).

*Conjecture 1 (The MDS Conjecture).* The parameters of a linear MDS code  $[n, k, n - k + 1]_q$  of length  $n$  and dimension  $k > 1$  over  $\mathbb{F}_q$  satisfy the following.

$$n \leq \begin{cases} q + 1 & \text{if } k \in \{2\} \cup \{4, 5, \dots, q - 2\}, \\ q + 1 & \text{if } k \in \{3, q - 1\} \text{ and } q \text{ is odd,} \\ q + 2 & \text{if } k \in \{3, q - 1\} \text{ and } q \text{ is even,} \\ k + 1 & \text{if } k \geq q. \end{cases}$$

A linear code corresponds to a row space of a  $k \times n$  matrix over  $\mathbb{F}_q$ . Such a matrix is called a *generator matrix* of the code. In particular, linear MDS codes can be uniquely represented by a  $k \times n$  matrix  $[I : M]$  which is a concatenation of  $k \times k$  identity matrix  $I$  and  $k \times (n - k)$  matrix  $M$  over  $\mathbb{F}_q$ . Here, the redundancy (check) part  $M$  of the generator matrix is called an *MDS matrix*. The minimum distance of a given linear MDS code equips the corresponding MDS matrix as in the following well-known result.

**Proposition 1.** *Let  $u$  be a nonzero  $k \times 1$  matrix and  $M$  be a  $k \times k$  MDS matrix over  $\mathbb{F}_q$ . Then the number of minimum total nonzero entries in both  $u$  and  $Mu$  is at least  $k + 1$ .*

Another important and characterizing feature of MDS matrices is the following algebraic property.

**Proposition 2.** *Let  $M$  be a  $k \times k$  matrix over  $\mathbb{F}_q$ . Then,  $M$  is MDS if and only if each submatrix of  $M$  is invertible.*

Construction of MDS matrices is an important and intensely studied area in mathematics and cryptography. We refer the reader to [GPR<sup>+</sup>19, Leu23] as recent and comprehensive surveys in this area.

## 2.2 Related Differentials

Daemen and Rijmen defined a new type of difference called related difference in [DR09]. They studied the propagation of these differences through the AES linear layer. They also defined related differentials, which are pairs of related differences combined by a matrix. We now use an extended version of their definition as follows.

**Definition 1.** [DR09] *Let  $M$  be an  $n \times n$  matrix over  $\mathbb{F}_q$ . Then, the pair of  $n$ -dimensional vectors  $x = (x_0, x_1, \dots, x_{n-1})$  and  $y = (y_0, y_1, \dots, y_{n-1})$  over  $\mathbb{F}_q$  is called a related differential if*

$$x_i y_i (x_i - y_i) = 0 \quad \text{and} \quad (Mx)_i (My)_i ((Mx)_i - (My)_i) = 0$$

for all  $i \in \{0, 1, \dots, n-1\}$ .

We express that, originally, the four-tuple  $(x, y, Mx, My)$  is called related differential in [BR22]. However, we prefer the notation  $(x, y)_M$  briefly, or only  $(x, y)$  more briefly if  $M$  is clear.

## 3 Existence of Related Differentials for $2 \times 2$ MDS Matrices

The following theorem solves the problem completely for all  $2 \times 2$  MDS matrices over any finite fields.

**Theorem 1.** *There are no related differentials for any  $2 \times 2$  MDS matrices.*

*Proof.* Let  $M$  be a  $2 \times 2$  MDS matrix over  $\mathbb{F}_q$ . Then

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

for some nonzero elements  $a, b, c, d$  in  $\mathbb{F}_q$  satisfying  $ad - bc \neq 0$  (i.e.  $\det M \neq 0$ ). On the other hand, we have only two types of related differentials

$$x = \begin{pmatrix} x_0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ y_1 \end{pmatrix}, \quad \text{and} \quad x = \begin{pmatrix} x_0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ y_1 \end{pmatrix}$$

without loss of generality, where  $x_0$  and  $y_1$  are nonzero elements in  $\mathbb{F}_q$ .

– In the first case, we have

$$Mx = \begin{pmatrix} ax_0 \\ cx_0 \end{pmatrix} \text{ and } My = \begin{pmatrix} by_1 \\ dy_1 \end{pmatrix}.$$

Clearly all entries of  $Mx$  and  $My$  are nonzero. Then we must have  $Mx = My$  to satisfy the zero differential property, which implies that  $\det M = 0$ , contradiction.

– In the second case, we have

$$Mx = \begin{pmatrix} ax_0 \\ cx_0 \end{pmatrix} \text{ and } My = \begin{pmatrix} ax_0 + by_1 \\ cx_0 + dy_1 \end{pmatrix}.$$

Clearly all entries of  $Mx$  are nonzero. Then we have  $(Mx)_0 = (My)_0$  or  $(Mx)_1 = (My)_1$ , then we obtain the contradictions  $by_1 = 0$  or  $dy_1 = 0$ , respectively.

As a result, we deduce that  $Mx$  and  $My$  cannot constitute a related differential.  $\square$

#### 4 Existence of Related Differentials for $3 \times 3$ Circulant MDS Matrices

One of the most popular MDS matrix types in the literature is circulant MDS matrices. In this section, we show that all such  $3 \times 3$  matrices have zero differentials as formulated in the following theorem.

**Theorem 2.** *There exist related differentials for all  $3 \times 3$  circulant MDS matrices.*

*Proof.* Let  $M$  be a  $3 \times 3$  circulant MDS matrix over  $\mathbb{F}_q$ . Then

$$M = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}$$

for some nonzero elements  $a, b, c$  in  $\mathbb{F}_q$  and any submatrices of  $M$  are invertible. Consider the pair

$$x = \begin{pmatrix} x_0 \\ -ab^{-1}x_0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ -ab^{-1}x_0 \\ cb^{-1}x_0 \end{pmatrix}$$

for some nonzero  $x_0 \in \mathbb{F}_q$ . Then we have

$$Mx = \begin{pmatrix} 0 \\ (c - a^2b^{-1})x_0 \\ (b - acb^{-1})x_0 \end{pmatrix} \text{ and } My = \begin{pmatrix} (c^2b^{-1})x_0 \\ (c - a^2b^{-1})x_0 \\ 0 \end{pmatrix}.$$

Clearly  $Mx$  and  $My$  satisfies the related differential property.  $\square$

*Remark 1.* We express that, among many  $(x, y)$  types satisfying the related differential property, i.e. among

$$\begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ y_1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ y_1 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} x_0 \\ y_1 \\ 0 \end{pmatrix}, \dots$$

the most suitable form is

$$\begin{pmatrix} x_0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ x_1 \\ y_2 \end{pmatrix}$$

for the proof of Theorem 2 according to our computations. For other types, either we have no related differentials or we need algebraic relations among  $a, b$ , and  $c$  some of which contradict with the fact that  $M$  is MDS. In other words, the density of related differentials among all possibilities is so low.

*Remark 2.* It is not surprising that there exist related differentials for all  $3 \times 3$  circulant MDS matrices. Recall that there exist related differentials also for all  $4 \times 4$  circulant MDS matrices as expressed in [DR09].

## 5 Existence of Related Differentials for $3 \times 3$ Involutory MDS Matrices over Finite Fields of Even Size

Involutory MDS matrices, whose square are the identity matrix, are a special class of MDS matrices. Note that, in case the implementation of the inverse of the MDS matrix is of interest (for decryption for example), involutory MDS matrices are more preferable since the efficiency and the implementation style of the matrix and its inverse are the same. Güzel et al. [GSA<sup>+</sup>19] presented the following characterization for  $3 \times 3$  involutory MDS matrices over finite fields whose characteristic is even.

**Proposition 3.** [GSA<sup>+</sup>19] *Any  $3 \times 3$  involutory MDS matrix over  $\mathbb{F}_{2^r}$  is in the form of*

$$\begin{pmatrix} a_1 & (a_1 + 1)b_0 & (a_1 + 1)b_1 \\ (a_2 + 1)b_0^{-1} & a_2 & (a_2 + 1)b_0^{-1}b_1 \\ (a_1 + a_2)b_1^{-1} & (a_1 + a_2)b_1^{-1}b_0 & a_1 + a_2 + 1 \end{pmatrix} \quad (1)$$

for some nonzero  $a_1, a_2, b_0, b_1 \in \mathbb{F}_{2^r}$  satisfying  $a_1 \neq a_2$ ,  $a_1 + a_2 \neq 1$ ,  $a_1 \neq 1$ , and  $a_2 \neq 1$ .

Now, we use the representation in Proposition 3 in order to prove the following theorem which is our main result in this section.

**Theorem 3.** *There exist related differentials for all  $3 \times 3$  involutory MDS matrices when the finite field has even characteristic.*

*Proof.* Assume that  $M$  is an MDS matrix in the form of Equation (1). Take

$$x = \begin{pmatrix} x_0 \\ b_0^{-1}x_0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ b_0^{-1}x_0 \\ b_1^{-1}x_0 \end{pmatrix}$$

for some nonzero  $x_0 \in \mathbb{F}_q$ . Then we have

$$Mx = \begin{pmatrix} x_0 \\ b_0^{-1}x_0 \\ 0 \end{pmatrix} \text{ and } My = \begin{pmatrix} 0 \\ b_0^{-1}x_0 \\ b_1^{-1}x_0 \end{pmatrix}$$

which clearly satisfies the related differential property.  $\square$

## 6 Characterization of All $3 \times 3$ MDS Matrices over Finite Fields of Even Size

In Section 4 and Section 5 we observe some negative results. We now search all  $3 \times 3$  matrices for positive results. For this purpose, the following reduction notion will be useful.

**Theorem 4.** *Let  $M$  be an arbitrary  $3 \times 3$  MDS matrix over finite field  $\mathbb{F}_q$ . Then there exist  $3 \times 3$  diagonal matrices  $D$  and  $E$  over  $\mathbb{F}_q$  such that  $M = DNE$ , where the first row and first column of  $N$  are all 1's. Moreover, there exist a related differential for  $M$  if and only if there exist a related differential for  $N$ .*

*Proof.* Let

$$M = \begin{pmatrix} m_1 & m_2 & m_3 \\ m_4 & m_5 & m_6 \\ m_7 & m_8 & m_9 \end{pmatrix}.$$

Clearly, we have  $m_i \neq 0$  for all  $i \in \{1, 2, \dots, 9\}$  since  $M$  is MDS. The first statement is trivial by

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & m_1 m_4^{-1} & 0 \\ 0 & 0 & m_1 m_7^{-1} \end{pmatrix} \quad \text{and} \quad E = \begin{pmatrix} m_1^{-1} & 0 & 0 \\ 0 & m_2^{-1} & 0 \\ 0 & 0 & m_3^{-1} \end{pmatrix}.$$

The second statement is immediate by the fact that  $E$  does not change the zero-ness and nonzero-ness property of the input differential  $(x, y)$ . Similarly,  $D$  does not change the zero-ness and nonzero-ness property of the output differential  $(Mx, My)$ .  $\square$

*Remark 3.* It is easy to extend Theorem 4 to any dimensions.

Now we can use the reduced form while investigating all  $3 \times 3$  MDS matrices when the field size is even.



**Theorem 5.** *Let  $M$  be a  $3 \times 3$  MDS matrix over  $\mathbb{F}_{2^r}$  and*

$$N = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & c & d \end{pmatrix}$$

*be its reduced form as described in Theorem 4 over  $\mathbb{F}_{2^r}$ . Then  $M$  has related differentials if and only if at least one of the following equations hold.*

- |                                     |                               |
|-------------------------------------|-------------------------------|
| i) $ad + b = 0$ ,                   | viii) $ad + bc + a + c = 0$ , |
| ii) $ad + c = 0$ ,                  | ix) $ad + bc + b + d = 0$ ,   |
| iii) $bc + a = 0$ ,                 | x) $ad + bc + c + d = 0$ ,    |
| iv) $bc + d = 0$ ,                  | xi) $ad + a + b + d = 0$ ,    |
| v) $a + b + c + d = 0$ ,            | xii) $ad + a + c + d = 0$ ,   |
| vi) $ad + bc + a + b + c + d = 0$ , | xiii) $bc + a + b + c = 0$ ,  |
| vii) $ad + bc + a + b = 0$ ,        | xiv) $bc + b + c + d = 0$ .   |

Before proving Theorem 5, we give an illustration by an infinite family of  $3 \times 3$  MDS matrices for which no related differentials exist.

*Example 1.* Let  $r > 8$  and  $\alpha$  be a primitive normal element of  $\mathbb{F}_{2^r}$  over  $\mathbb{F}_2$ . Let  $D$  and  $E$  be two arbitrary invertible diagonal  $3 \times 3$  matrices over  $\mathbb{F}_{2^r}$ . Let

$$N = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^{2^2} & \alpha^{2^3} \end{pmatrix}.$$

Then  $DNE$  is an MDS matrix for which there exist no related differentials. It is clear that none of the 14 equations is valid for  $a = \alpha$ ,  $b = \alpha^2$ ,  $c = \alpha^4$ , and  $d = \alpha^8$ .

We now prove Theorem 5 as follows.

*Proof.* Since  $M$  is an MDS matrix, we have these conditions:  $ad - bc \neq 0$ ,  $a \neq b$ ,  $a \neq c$ ,  $d \neq b$ ,  $d \neq c$ , and  $a, b, c, d \notin \{0, 1\}$ . Let  $x$  and  $y$  be two related differences, and  $x' = Mx$  and  $y' = My$  be output differences. If  $x'$  and  $y'$  are related differences, we can conclude  $M$  has related differential.

We list all input differences which are related and check whether the output pairs are related differences or not.

**Case 1:** Assume that  $x$  and  $y$  have one non-zero entry.

$$\begin{aligned} \text{i) } x &= \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ y_1 \\ 0 \end{pmatrix} \\ \text{So, } x' &= \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}, y' = \begin{pmatrix} y_1 \\ ay_1 \\ cy_1 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + y_1 \\ x_0 + ay_1 \\ x_0 + cy_1 \end{pmatrix}. \end{aligned}$$

Since  $x_0 \neq 0$  and  $y_1 \neq 0$ ,  $x' \oplus y' = 0$ . Therefore,  $x_0 + y_1 = 0$  which means  $x_0 = y_1$ . Also  $x_0 + ay_1 = 0$ ,  $x_0 + cy_1 = 0$ . This implies  $a$  and  $c$  are 1 which is a contradiction. Thus there is no related differences in this case.

$$\text{ii) } x = \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ 0 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}, y' = \begin{pmatrix} y_2 \\ by_2 \\ dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + y_2 \\ x_0 + by_2 \\ x_0 + dy_2 \end{pmatrix}.$$

Because of very similar reason to i, we get  $b = 1$  and  $d = 1$ . So this is a contradiction.

$$\text{iii) } x = \begin{pmatrix} 0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ 0 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix}, y' = \begin{pmatrix} y_2 \\ by_2 \\ dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_1 + y_2 \\ ax_1 + by_2 \\ cx_1 + dy_2 \end{pmatrix}.$$

Since  $x_1$  and  $y_2$  are nonzero,  $x' \oplus y' = 0$ . We have  $x_1 + y_2 = 0$ ,  $ax_1 + by_2 = 0$ ,  $cx_1 + dy_2 = 0$ . That means  $a = b$  and  $c = d$  which are contradictions.

**Case 2:**  $x$  has one and  $y$  has two non-zero entries.

$$\text{i) } x = \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ y_1 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}, y' = \begin{pmatrix} x_0 + y_1 \\ x_0 + ay_1 \\ x_0 + cy_1 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_1 \\ ay_1 \\ cy_1 \end{pmatrix}.$$

This case is the same with Case 1-i.

$$\text{ii) } x = \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ 0 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}, y' = \begin{pmatrix} x_0 + y_2 \\ x_0 + by_2 \\ x_0 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_2 \\ by_2 \\ dy_2 \end{pmatrix}.$$

This case is the same with Case 1-ii.

$$\text{iii) } x = \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ y_1 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}, y' = \begin{pmatrix} y_1 + y_2 \\ ay_1 + by_2 \\ cy_1 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + y_1 + y_2 \\ x_0 + ay_1 + by_2 \\ x_0 + cy_1 + dy_2 \end{pmatrix}.$$

Since  $x_0, y_1$  and  $y_2$  are non-zero, we have two possibilities:  $y_1 + y_2 = 0$ ;  $x_0 + y_1 + y_2 = 0$ .

Firstly, assume  $y_1 + y_2 = 0$  which means  $y_1 = y_2$ . Then for the second entry we have  $ay_1 + by_2 = 0$  or  $x_0 + ay_1 + by_2 = 0$ . If  $ay_1 + by_2 = 0 \rightarrow (a + b)y_1 = 0 \rightarrow a = b$ , it is impossible. So  $x_0 + ay_1 + by_2 = 0 \rightarrow$

$x_0 + (a + b)y_1 = 0$ . For the third entry, we have  $cy_1 + dy_2 = 0$  which results to  $c = d$  so it a contradiction. Therefore,  $x_0 + cy_1 + dy_2 = 0 \rightarrow (a + b)y_1 + cy_1 + dy_1 = 0$ . Thus we get **a+b+c+d=0**.

Secondly, assume  $x_0 + y_1 + y_2 = 0$ . For the second entry we have  $ay_1 + by_2 = 0$  or  $x_0 + ay_1 + by_2 = 0$ . If  $ay_1 + by_2 = 0 \rightarrow y_2 = \frac{a}{b}y_1$ . In this case, for the third entry, again we have two options:  $cy_1 + dy_2 = 0 \rightarrow cy_1 + \frac{ad}{b}y_1 = 0 \rightarrow ad + bc = 0$  which is a contradiction or  $x_0 + cy_1 + dy_2 = 0 \rightarrow (c + 1)y_1 + (d + 1)y_2 = 0 \rightarrow (c + 1)y_1 + (d + 1)\frac{a}{b}y_1 = 0$ . Thus we get **ad+bc+a+b=0**. On the other hand, if  $x_0 + ay_1 + by_2 = 0 \rightarrow (a + 1)y_1 + (b + 1)y_2 = 0$ . For the third entry, if  $cy_1 + dy_2 = 0$ , then  $cy_1 + d\frac{a+1}{b+1}y_1 = 0$  so we get **ad+bc+c+d=0**. Lastly, if  $x_0 + cy_1 + dy_2 = 0$ , then  $(c + 1)y_1 + (d + 1)y_2 = 0 \rightarrow (c + 1)y_1 + (d + 1)\frac{a+1}{b+1}y_1 = 0$ . In this case we get **ad+bc+a+b+c+d=0**.

$$\text{iv) } x = \begin{pmatrix} 0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} y_0 \\ x_1 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix}, y' = \begin{pmatrix} x_1 + y_0 \\ ax_1 + y_0 \\ cx_1 + y_0 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_0 \\ y_0 \\ y_0 \end{pmatrix}.$$

Since  $x_1$  and  $y_0$  are nonzero,  $y' = 0$ . We have  $x_1 + y_0 = 0$ ,  $ax_1 + y_0 = 0$ ,  $cx_1 + y_0 = 0$ . That means  $x_1 = y_0$  and so  $a = c = 1$  which are contradictions.

$$\text{v) } x = \begin{pmatrix} 0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ x_1 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix}, y' = \begin{pmatrix} x_1 + y_2 \\ ax_1 + by_2 \\ cx_1 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_2 \\ by_2 \\ dy_2 \end{pmatrix}.$$

This is the same with Case 1-iii.

$$\text{vi) } x = \begin{pmatrix} 0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} y_0 \\ 0 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix}, y' = \begin{pmatrix} y_0 + y_2 \\ y_0 + by_2 \\ y_0 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_1 + y_0 + y_2 \\ ax_1 + y_0 + by_2 \\ cx_1 + y_0 + dy_2 \end{pmatrix}.$$

Since all input values are non-zero, we have two possibilities:  $y_0 + y_2 = 0$  or  $x_1 + y_0 + y_2 = 0$ .

Firstly, assume  $y_0 + y_2 = 0 \rightarrow y_0 = y_2$ . For the second entry  $y_0 + by_2 = 0$  or  $ax_1 + y_0 + by_2 = 0$ . If  $y_0 + by_2 = 0$ ,  $(b + 1)y_0 = 0 \rightarrow b = 1$  which is not possible. If  $ax_1 + y_0 + by_2 = 0$ , we get  $ax_1 + (b + 1)y_0 = 0 \rightarrow x_1 = \frac{b+1}{a}y_0$ . For the third entry again  $y_0 + dy_2 = 0 \rightarrow d = 1$  which results to contradiction and  $cx_1 + y_0 + dy_2 = 0 \rightarrow cx_1 + (d + 1)y_0 = 0 \rightarrow (c\frac{b+1}{a} + (d + 1))y_0 = 0$  Therefore, we get **ad+bc+a+c=0**.

Secondly, assume  $x_1 + y_0 + y_2 = 0$  for the first row and the second one again we have two options. One of them is  $y_0 + by_2 = 0 \rightarrow y_0 = by_2$ . In

this case for the third row, we have  $y_0 + dy_2 = 0$  which is a contradiction because  $(b+d)y_2 = 0 \rightarrow b = d$  or  $cx_1 + y_0 + dy_2 = 0 \rightarrow cx_1 + (b+d)y_2 = 0 \rightarrow c(y_0 + y_2) + (b+d)y_2 = 0 \rightarrow bcy_2 + cy_2 + (b+d)y_2 = 0$ . Thus, we get **bc+c+b+d=0**.

The other case for the second row is  $ax_1 + y_0 + by_2 = 0 \rightarrow (a+1)x_1 + (b+1)y_2 = 0$ . For the third row one case is  $y_0 + dy_2 = 0$  and the other is  $cx_1 + y_0 + dy_2 = 0$ . If  $y_0 + dy_2 = 0 \rightarrow x_1 + (d+1)y_2 = (\frac{b+1}{a+1} + (d+1))y_2 = 0$ . Thus, we get **ad+a+b+d=0**.

For the  $cx_1 + y_0 + dy_2 = 0$ , we have  $(c+1)x_1 + (d+1)y_2 = 0 \rightarrow ((c+1)\frac{b+1}{a+1} + (d+1))y_2 = 0$ . Thus, we get **ad+bc+a+b+c+d=0**.

$$\text{vii) } x = \begin{pmatrix} 0 \\ 0 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} y_0 \\ 0 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_2 \\ bx_2 \\ dx_2 \end{pmatrix}, y' = \begin{pmatrix} y_0 + x_2 \\ y_0 + bx_2 \\ y_0 + dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_0 \\ y_0 \\ y_0 \end{pmatrix}.$$

Since  $x_2 \neq 0$  and  $y_0 \neq 0$ , only  $y'$  can be zero. So  $y_0 + x_2 = 0 \rightarrow y_0 = x_2$ . This means  $b = 1$  and  $d = 1$  for the other rows which is a contradiction.

$$\text{viii) } x = \begin{pmatrix} 0 \\ 0 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ y_1 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_2 \\ bx_2 \\ dx_2 \end{pmatrix}, y' = \begin{pmatrix} y_1 + x_2 \\ ay_1 + bx_2 \\ cy_1 + dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} y_1 \\ ay_1 \\ cy_1 \end{pmatrix}.$$

Since  $x_2 \neq 0$  and  $y_1 \neq 0$ , only  $y'$  can be zero. So  $y_1 + x_2 = 0 \rightarrow y_1 = x_2$ . This means  $a = b$  and  $c = d$  for the other rows which is a contradiction.

$$\text{ix) } x = \begin{pmatrix} 0 \\ 0 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} y_0 \\ y_1 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_2 \\ bx_2 \\ dx_2 \end{pmatrix}, y' = \begin{pmatrix} y_0 + y_1 \\ y_0 + ay_1 \\ y_0 + cy_1 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_2 + y_0 + y_1 \\ bx_2 + y_0 + ay_1 \\ dx_2 + y_0 + cy_1 \end{pmatrix}.$$

For the first row, either  $y_0 + y_1 = 0$  or  $x_2 + y_0 + y_1 = 0$

If  $y_0 + y_1 = 0$ , we get  $y_0 = y_1$ . In the second row, either  $y_0 + ay_1 = 0$  or  $bx_2 + y_0 + ay_1 = 0$ . If  $y_0 + ay_1 = 0$ ,  $y_0 + ay_0 = 0 \rightarrow a = 1$  which is a contradiction. So  $bx_2 + y_0 + ay_1 = 0$  and we get  $bx_2 + (a+1)y_0 = 0 \rightarrow x_2 = \frac{a+1}{b}y_0$ . And for the last row, if  $y_0 + cy_1 = 0$ , again we get a contradiction because of  $c = 1$ . Therefore,  $dx_2 + y_0 + cy_1 = 0$ . In this case,  $dx_2 + (c+1)y_0 = 0 \rightarrow (\frac{ad+d}{b} + c + 1)y_0 = 0$ . Thus, we get **ad+bc+b+d=0**.

Now we study the other case for the first row:  $x_2 + y_0 + y_1 = 0$ . For the second row again we have either  $y_0 + ay_1 = 0$  or  $bx_2 + y_0 + ay_1 = 0$ . If  $y_0 + ay_1 = 0$ , then we can say  $y_0 = ay_1$ . For the third row,

$y_0 + cy_1 = 0 \rightarrow (a + c)y_1 = 0 \rightarrow a = c$  we get contradiction. So the only way is  $dx_2 + y_0 + cy_1 = 0$ . Then  $d(y_0 + y_1) + y_0 + cy_1 = 0 \rightarrow d(ay_1 + y_1) + (a + c)y_1 = 0 \rightarrow \mathbf{ad} + \mathbf{a} + \mathbf{c} + \mathbf{d} = \mathbf{0}$ .

If the second row is  $bx_2 + y_0 + ay_1 = 0$  we can say  $(b + 1)x_2 = (a + 1)y_1$ . For the third row, if  $y_0 + cy_1 = 0 \rightarrow bx_2 + ay_1 + cy_1 = 0 \rightarrow (b\frac{a+1}{b+1} + a + c)y_1 = 0$ . Therefore we get  $\mathbf{bc} + \mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$ .

The last property is the third row  $dx_2 + y_0 + cy_1 = 0$ . Then  $(d + 1)x_2 + (c + 1)y_1 = 0 \rightarrow (d + 1)x_2 + (c + 1)y_1 = 0 \rightarrow (d + 1)\frac{a+1}{b+1}y_1 + (c + 1)y_1 = 0$ . Thus, we get  $\mathbf{ad} + \mathbf{bc} + \mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \mathbf{0}$ .

**Case 3:** Both of  $x$  and  $y$  have two non-zero entries.

$$\text{i) } x = \begin{pmatrix} x_0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ 0 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 \\ x_0 + ax_1 \\ x_0 + cx_1 \end{pmatrix}, y' = \begin{pmatrix} x_0 + y_2 \\ x_0 + by_2 \\ x_0 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_1 + y_2 \\ ax_1 + by_2 \\ cx_1 + dy_2 \end{pmatrix}.$$

Since this case is the same with the following one, we can skip this.

$$\text{ii) } x = \begin{pmatrix} x_0 \\ x_1 \\ 0 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ x_1 \\ y_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 \\ x_0 + ax_1 \\ x_0 + cx_1 \end{pmatrix}, y' = \begin{pmatrix} x_1 + y_2 \\ ax_1 + by_2 \\ cx_1 + dy_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + y_2 \\ x_0 + by_2 \\ x_0 + dy_2 \end{pmatrix}.$$

In this case, we have three options for the first row:  $x_0 + x_1 = 0$ ,  $x_1 + y_2 = 0$  or  $x_0 + y_2 = 0$

If  $x_0 + x_1 = 0 \rightarrow x_0 = x_1$ , then for the second row again either  $x_0 + ax_1 = 0$  or  $ax_1 + by_2 = 0$  or  $x_0 + by_2 = 0$ . If  $x_0 + ax_1 = 0 \rightarrow (a + 1)x_1 = 0 \rightarrow a = 1$  which is a contradiction. If  $ax_1 + by_2 = 0$ , then  $x_1 = \frac{b}{a}y_2$ . For the last row,  $x_0 + cx_1 = 0$  means  $c = 1$  again we get a contradiction. If  $cx_1 + dy_2 = 0 \rightarrow c\frac{b}{a}y_2 + dy_2 = 0$ , we get  $bc + ad = 0$  which is a contradiction. So the only option for the last row is  $x_0 + dy_2 = 0 \rightarrow x_1 + dy_2 = 0 \rightarrow \frac{b}{a}y_2 + dy_2 = 0$ . Thus, we get  $\mathbf{ad} + \mathbf{b} = \mathbf{0}$ . If the second is  $x_0 + by_2 = 0$ , we get contradiction for the  $x_0 + cx_1 = 0 \rightarrow c = 1$ . If  $cx_1 + dy_2 = 0 \rightarrow cx_0 + dy_2 = 0 \rightarrow cby_2 + dy_2 = 0$ , we get  $\mathbf{bc} + \mathbf{d} = \mathbf{0}$ . If  $x_0 + dy_2 = 0 \rightarrow by_2 + dy_2 = 0$ , we get the contradiction due to  $b \neq d$ .

If  $x_1 + y_2 = 0 \rightarrow x_1 = y_2$ , then again for the second row, either  $x_0 + ax_1 = 0$  or  $ax_1 + by_2 = 0$  or  $x_0 + by_2 = 0$ . If  $x_0 + ax_1 = 0 \rightarrow x_0 = ax_1 = ay_2$ . In this case for the last row,  $x_0 + cx_1 = 0 \rightarrow a = c$  and  $cx_1 + dy_2 = 0 \rightarrow c = d$  result to contradictions. If  $x_0 + dy_2 = 0 \rightarrow ay_2 + dy_2 = 0$ , we get  $\mathbf{a} = \mathbf{d}$ . If  $ax_1 + by_2 = 0 \rightarrow ax_1 + bx_1 = 0 \rightarrow a = b$ , we get the contradiction. The last case for the second row is  $x_0 + by_2 = 0$  which means  $x_0 = bx_1 = by_2$ . For the last row, in cases  $cx_1 + dy_2 = 0 \rightarrow (c + d)y_2 = 0 \rightarrow c = d$  and  $x_0 + dy_2 = 0 \rightarrow by_2 + dy_2 = 0 \rightarrow b = d$ , we get contradictions. However, in  $x_0 + cx_1 = 0 \rightarrow bx_1 + cx_1 = 0$ , we get  $\mathbf{b} = \mathbf{c}$ .

If  $x_0 + y_2 = 0 \rightarrow x_0 = y_2$ , then again for the second row, either  $x_0 + ax_1 = 0$  or  $ax_1 + by_2 = 0$  or  $x_0 + by_2 = 0$ . The last one is impossible because in this case we get  $b = 1$ . If  $x_0 + ax_1 = 0 \rightarrow x_0 = ax_1 = y_2$ . For the last row,  $x_0 + cx_1 = 0 \rightarrow a = c$  and  $x_0 + dy_2 = 0 \rightarrow d = 1$  are contradictions. If  $cx_1 + dy_2 = 0 \rightarrow cx_1 + dax_1 = 0$ , we get **ad+c=0**. If  $ax_1 + by_2 = 0 \rightarrow x_0 = \frac{a}{b}x_1 = y_2$ . For the last row,  $cx_1 + dy_2 = 0 \rightarrow cx_1 + d\frac{a}{b}x_1 = 0 \rightarrow ad + bc = 0$  and  $x_0 + dy_2 = 0 \rightarrow d = 1$  are contradictions. If  $x_0 + cx_1 = 0 \rightarrow \frac{a}{b}x_1 + cx_1 = 0$ , we get **bc+a=0**.

$$\text{iii) } x = \begin{pmatrix} x_0 \\ 0 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ y_1 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_2 \\ x_0 + bx_2 \\ x_0 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_2 + y_1 \\ ay_1 + bx_2 \\ cy_1 + dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + y_1 \\ x_0 + ay_1 \\ x_0 + cy_1 \end{pmatrix}.$$

In this case, we have three options for the first row:  $x_0 + x_2 = 0$ ,  $x_2 + y_1 = 0$  or  $x_0 + y_1 = 0$

Firstly, we study  $x_0 + x_2 = 0 \rightarrow x_0 = x_2$ . For the second row,  $x_0 + bx_2 = x_2 + bx_2 = 0$  is the contradiction due to  $b = 1$ . If  $ay_1 + bx_2 = 0 \rightarrow x_0 = x_2 = \frac{a}{b}y_1$ , we get contradictions in cases  $x_0 + dx_2 = 0 \rightarrow d = 1$  and  $cy_1 + dx_2 = 0 \rightarrow cy_1 + d\frac{a}{b}y_1 = 0 \rightarrow ad + bc = 1$ . Therefore,  $x_0 + cy_1 = 0$  is the only possibility for the third row and we get  $x_0 + cy_1 = 0 \rightarrow \frac{a}{b}y_1 + cy_1 = 0 \rightarrow \mathbf{bc+a=0}$ . On the other hand, if  $x_0 + ay_1 = 0 \rightarrow x_0 = x_2 = ay_1$ , the only way for third row is  $cy_1 + dx_2 = 0 \rightarrow cy_1 + day_1 = 0$  and we get **ad+c=0**.

Secondly, if  $x_2 + y_1 = 0 \rightarrow x_2 = y_1$ ,  $ay_1 + bx_2 = 0$  is contradiction due to  $a \neq b$ . If  $x_0 + bx_2 = 0 \rightarrow x_0 = bx_2 = by_1$ ,  $x_0 + dx_2 = 0 \rightarrow b = d$  and  $cy_1 + dx_2 = 0 \rightarrow c = d$  are contradictions. Therefore, the only way for the last row is  $x_0 + cy_1 = 0 \rightarrow by_1 + cy_1 = 0$  and we get **b=c**. Similarly,  $cy_1 + dx_2 = 0$  and  $x_0 + cy_1 = 0$  are impossible in case of  $x_0 + ay_1 = 0$ . In this case the only way is  $x_0 + dx_2 = 0 \rightarrow ax_2 + dx_2 = 0$  and we get **a=d**.

Thirdly, if  $x_0 + y_1 = 0 \rightarrow x_0 = y_1$ ,  $x_0 + ay_1 = 0$  is contradiction due to  $a \neq 1$ . In case  $x_0 + bx_2 = 0 \rightarrow x_0 = y_1 = bx_2$ , if  $x_0 + dx_2 = 0 \rightarrow bx_2 + dx_2 = 0 \rightarrow b = d$  and  $x_0 + cy_1 = 0 \rightarrow x_0 + cx_0 = 0 \rightarrow c = 1$ , we get contradictions. If  $cy_1 + dx_2 = 0 \rightarrow cbx_2 + dx_2 = 0$ , we get **bc+d=0**. On the other hand, if  $ay_1 + bx_2 = 0 \rightarrow x_0 = y_1 = \frac{b}{a}x_2$ ,  $cy_1 + dx_2 = 0 \rightarrow c\frac{b}{a}x_2 + dx_2 = 0 \rightarrow bc + ad = 0$  and  $x_0 + cy_1 = 0 \rightarrow c = 1$  are contradictions but if  $x_0 + dx_2 = 0 \rightarrow \frac{b}{a}x_2 + dx_2 = 0$ , we get we get **ad+b=0**.

**Case 4:** x has 3 and y has one non-zero entries.

$$\text{i) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_1 + x_2 \\ ax_1 + bx_2 \\ cx_1 + dx_2 \end{pmatrix}.$$

For the first row, we have two options, either  $x_0 + x_1 + x_2 = 0$  or  $x_1 + x_2 = 0$

If  $x_1 + x_2 = 0 \rightarrow x_1 = x_2$ , we get a contradiction for case  $ax_1 + bx_2 = 0 \rightarrow (a+b)x_1 = 0 \rightarrow a = b$ . On the other hand, in the case  $x_0 + ax_1 + bx_2 = 0 \rightarrow x_0 = (a+b)x_1$ , we have only one option because  $cx_1 + dx_2 = 0 \rightarrow c = d$  is a contradiction, so we have  $x_0 + cx_1 + dx_2 = 0 \rightarrow x_0 + (c+d)x_1 = 0 \rightarrow (a+b+c+d)x_1 = 0$  and **a+b+c+d=0**.

Now let  $x_0 + x_1 + x_2 = 0$ . If  $x_0 + ax_1 + bx_2 = 0 \rightarrow (a+1)x_1 + (b+1)x_2 = 0 \rightarrow x_2 = \frac{a+1}{b+1}x_1$ . In this case, we have two options for the last row. Firstly, if  $x_0 + cx_1 + dx_2 = 0 \rightarrow (c+1)x_1 + (d+1)x_2 = 0 \rightarrow ((c+1) + (d+1)\frac{a+1}{b+1})x_1 = 0$ , we get **ad+bc+a+b+c+d=0**. Secondly, if  $cx_1 + dx_2 = 0 \rightarrow cx_1 + d\frac{a+1}{b+1}x_1 = 0$ , we get **ad+bc+c+d=0**.

In this case, if we assume  $ax_1 + bx_2 = 0$  for the second row, we have only one option because  $cx_1 + dx_2 = 0 \rightarrow cx_1 + d\frac{a}{b}x_1 = 0 \rightarrow ad+bc = 0$  is the contradiction. The possible case is  $x_0 + cx_1 + dx_2 = 0$  and we get **ad+bc+a+b=0**.

$$\text{ii) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ x_1 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + x_2 \\ x_0 + bx_2 \\ x_0 + dx_2 \end{pmatrix}.$$

For the first row, we have two options, either  $x_0 + x_2 = 0$  or  $x_0 + x_1 + x_2 = 0$ .

Let  $x_0 + x_2 = 0 \rightarrow x_0 = x_2$ . For  $x' \oplus y'$ , we get the contradiction in the second and third row. Therefore, the second row can be  $x_0 + ax_1 + bx_2 = 0 \rightarrow x_1 = \frac{b+1}{a}x_0$ . For the third row,  $x_0 + cx_1 + dx_2 = 0$ . Then  $(b+1)x_0 + cx_1 = 0 \rightarrow (d+1)x_0 + c\frac{b+1}{a}x_0 = 0$ . Thus, we get **ad+bc+a+c=0**.

Now let  $x_0 + x_1 + x_2 = 0$ . If  $x_0 + ax_1 + bx_2 = 0 \rightarrow (a+1)x_1 + (b+1)x_2 = 0$ , we have two different cases for the last row. The first is  $x_0 + cx_1 + dx_2 = 0$ . In this case,  $(c+1)x_1 + (d+1)x_2 = 0 \rightarrow (c+1)x_1 + (d+1)\frac{a+1}{b+1}x_1 = 0$  and we get **ad+bc+a+b+c+d=0**. The second is  $x_0 + dx_2 = 0 \rightarrow x_1 + (d+1)x_2 = 0 \rightarrow \frac{b+1}{a+1}x_2 + (d+1)x_2 = 0$ , we get **ad+a+b+d=0**. If we change the case for the second row that is  $x_0 + bx_2 = 0 \rightarrow x_0 = bx_2$ , we get the contradiction in case  $x_0 + dx_2 = bx_2 + dx_2 = 0 \rightarrow b = d$  but in case  $x_0 + cx_1 + dx_2 = x_0 + c(x_0 + x_2) + dx_2 = (c+1)bx_2 + (c+d)x_2$ , we get **bc+b+c+d=0**.

$$\text{iii) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ 0 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_2 \\ bx_2 \\ dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 + x_1 \\ x_0 + ax_1 \\ x_0 + cx_1 \end{pmatrix}.$$

For the first row, we have two options, either  $x_0 + x_1 = 0$  or  $x_0 + x_1 + x_2 = 0$ .

Similar to the previous cases, we get  $\mathbf{ad} + \mathbf{bc} + \mathbf{b} + \mathbf{d} = \mathbf{0}$  if  $x_0 + x_1 = 0$ ,  $x_0 + ax_1 + bx_2$  and  $x_0 + cx_1 + dx_2 = 0$ . We get some contradictions in other cases when  $x_0 + x_1 = 0$ .

On the other hand, if  $x_0 + x_1 + x_2 = 0$ , we get  $\mathbf{ad} + \mathbf{bc} + \mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \mathbf{0}$  in case  $x_0 + ax_1 + bx_2 = 0$  and  $x_0 + cx_1 + dx_2 = 0$ . We get  $\mathbf{bc} + \mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$  in case  $x_0 + ax_1 + bx_2 = 0$  and  $x_0 + cx_1 = 0$ . Lastly, we get  $\mathbf{ad} + \mathbf{a} + \mathbf{c} + \mathbf{d} = \mathbf{0}$  in case  $x_0 + ax_1 = 0$  and  $x_0 + cx_1 + dx_2 = 0$ .

**Case 5:**  $x$  has 3 and  $y$  has two non-zero entries.

$$\text{i) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ x_1 \\ 0 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_0 + x_1 \\ x_0 + ax_1 \\ x_0 + cx_1 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_2 \\ bx_2 \\ dx_2 \end{pmatrix}.$$

This is the same case with 4-iii.

$$\text{ii) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} x_0 \\ 0 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_0 + x_2 \\ x_0 + bx_2 \\ x_0 + dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_1 \\ ax_1 \\ cx_1 \end{pmatrix}.$$

This is the same case as 4-ii.

$$\text{iii) } x = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} 0 \\ x_1 \\ x_2 \end{pmatrix}$$

$$\text{So, } x' = \begin{pmatrix} x_0 + x_1 + x_2 \\ x_0 + ax_1 + bx_2 \\ x_0 + cx_1 + dx_2 \end{pmatrix}, y' = \begin{pmatrix} x_1 + x_2 \\ ax_1 + bx_2 \\ cx_1 + dx_2 \end{pmatrix} \text{ and } x' \oplus y' = \begin{pmatrix} x_0 \\ x_0 \\ x_0 \end{pmatrix}.$$

This is the same case with 4-i.

□

## 7 Conclusion and Future Work

In this paper, we prove the existence and nonexistence of related differentials for MDS matrices over finite fields in several cases, in particular:

- There are no related differentials for any  $2 \times 2$  MDS matrices.
- There exist related differentials for all  $3 \times 3$  circulant MDS matrices.



- There exist related differentials for all  $3 \times 3$  involutory MDS matrices when the finite field has even size.
- We characterize all  $3 \times 3$  MDS matrices for which there exist no related differentials when the finite field has even size.

In that way, in various cases, we observe explicitly which types of MDS matrices have more/less cryptographic resilience.

Note that it is possible to extend our results in various directions, since there are several natural questions arising within this scope, for example:

1. For  $4 \times 4$  MDS matrices, we have answers for circulant and Hadamard matrices so far. So, what can we say for other types of  $4 \times 4$  MDS matrices; particularly for generalized Hadamard, Cauchy, involutory matrices?
2. What happens for  $n \times n$  MDS matrices when  $n \geq 5$ ?
3. What can we say for non-MDS matrices? For example, what happens for near-MDS matrices? Binary matrices?
4. The number and types of related differentials is another interesting question. Note that some related differentials may be more easy to exploit in zero-difference attacks.

## References

- [AES01] Advanced Encryption Standard (AES). National Institute of Standards and Technology, NIST FIPS PUB 197, U.S. Department of Commerce, November 2001.
- [AMPŠ24] Tomer Ashur, Mohammad Mahzoun, Jim Posen, and Danilo Šijačić. Vision mark-32: Zk-friendly hash function over binary tower fields. *Cryptology ePrint Archive*, 2024.
- [BLNS18] Christina Boura, Virginie Lallemand, María Naya-Plasencia, and Valentin Suder. Making the impossible possible. *Journal of Cryptology*, 31(1):101–133, January 2018.
- [BNN<sup>+</sup>10] Paulo S. L. M. Barreto, Ventislav Nikov, Svetla Nikova, Vincent Rijmen, and Elmar Tischhauser. Whirlwind: A new cryptographic hash function. *Designs, Codes and Cryptography*, 56(2–3):141–162, 2010.
- [BR22] Navid Ghaedi Bardeh and Vincent Rijmen. New key-recovery attack on reduced-round AES. *IACR Trans. Symm. Cryptol.*, 2022(2):43–62, 2022.
- [BS91] Eli Biham and Adi Shamir. Differential cryptanalysis of DES-like cryptosystems. *Journal of Cryptology*, 4(1):3–72, January 1991.
- [DFJ13] Patrick Derbez, Pierre-Alain Fouque, and Jérémy Jean. Improved key recovery attacks on reduced-round AES in the single-key setting. In Thomas Johansson and Phong Q. Nguyen, editors, *EUROCRYPT 2013*, volume 7881 of *LNCS*, pages 371–387. Springer, Heidelberg, May 2013.
- [DKS10] Orr Dunkelman, Nathan Keller, and Adi Shamir. Improved single-key attacks on 8-round AES-192 and AES-256. In Masayuki Abe, editor, *ASIACRYPT 2010*, volume 6477 of *LNCS*, pages 158–176. Springer, Heidelberg, December 2010.
- [DR09] Joan Daemen and Vincent Rijmen. New criteria for linear maps in aes-like ciphers. *Cryptogr. Commun.*, 1(1):47–69, 2009.

- [DS08] Hüseyin Demirci and Ali Aydin Selçuk. A meet-in-the-middle attack on 8-round AES. In Kaisa Nyberg, editor, *FSE 2008*, volume 5086 of *LNCS*, pages 116–126. Springer, Heidelberg, February 2008.
- [G<sup>+</sup>21] Lorenzo Grassi et al. Poseidon: A new hash function for zero-knowledge proof systems. In *30th USENIX Security Symposium*, pages 519–535. USENIX Association, 2021.
- [GPR<sup>+</sup>19] Kishan Chand Gupta, Sumit Kumar Pandey, Indranil Ghosh Ray, , and Susanta Samanta. Cryptographically significant MDS matrices over finite fields: a brief survey and some generalized results. *Advances in Mathematics of Communications*, 13(4):779–843, 2019.
- [GSA<sup>+</sup>19] Gülsüm Gözde Güzel, Muharrem Tolga Sakalli, Sedat Akleylek, Vincent Rijmen, and Yasemin Cengellenmis. A new matrix form to generate all  $3 \times 3$  involutory MDS matrices over  $\mathbb{F}_2^m$ . *Inf. Process. Lett.*, 147:61–68, 2019.
- [JLG25] Sonu Jha, Shun Li, and Danilo Gligoroski. Construction of hadamard-based mixcolumns matrices resistant to related-differential cryptanalysis. *IACR Commun. Cryptol.*, 2(1):37, 2025.
- [LDKK08] Jiqiang Lu, Orr Dunkelman, Nathan Keller, and Jongsung Kim. New impossible differential attacks on AES. In Dipanwita Roy Chowdhury, Vincent Rijmen, and Abhijit Das, editors, *INDOCRYPT 2008*, volume 5365 of *LNCS*, pages 279–293. Springer, Heidelberg, December 2008.
- [Leu23] Gaëtan Leurent. MDS matrices. In *Symmetric Cryptography 1: Design and Security Proofs*, pages 99–109. Wiley Online Library, 2023.
- [LP21] Gaëtan Leurent and Clara Pernot. New representations of the AES key schedule. In Anne Canteaut and François-Xavier Standaert, editors, *EUROCRYPT 2021, Part I*, volume 12696 of *LNCS*, pages 54–84. Springer, Heidelberg, October 2021.
- [Mat94] Mitsuru Matsui. Linear cryptanalysis method for DES cipher. In Tor Helleseth, editor, *EUROCRYPT'93*, volume 765 of *LNCS*, pages 386–397. Springer, Heidelberg, May 1994.
- [RBH17] Sondre Rønjom, Navid Ghaedi Bardeh, and Tor Helleseth. Yoyo tricks with AES. In Tsuyoshi Takagi and Thomas Peyrin, editors, *ASIACRYPT 2017, Part I*, volume 10624 of *LNCS*, pages 217–243. Springer, Heidelberg, December 2017.
- [Rot06] Ron M. Roth. Introduction to Coding Theory. Cambridge, UK: Cambridge University Press, 2006.
- [SLS<sup>+</sup>23] Alan Szepieniec, Alexander Lemmens, Jan Ferdinand Sauer, Bobbin Threadbare, and Al-Kindi. The tip5 hash function for recursive STARKs. Cryptology ePrint Archive, Paper 2023/107, 2023.